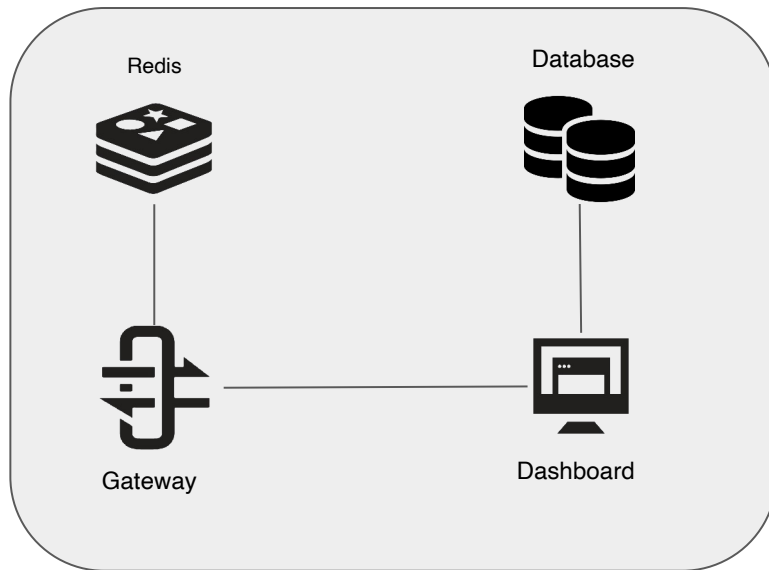


Tyk Database: MongoDB & PostgreSQL

Understanding the role of Tyk's primary databases and how they support performance and high availability

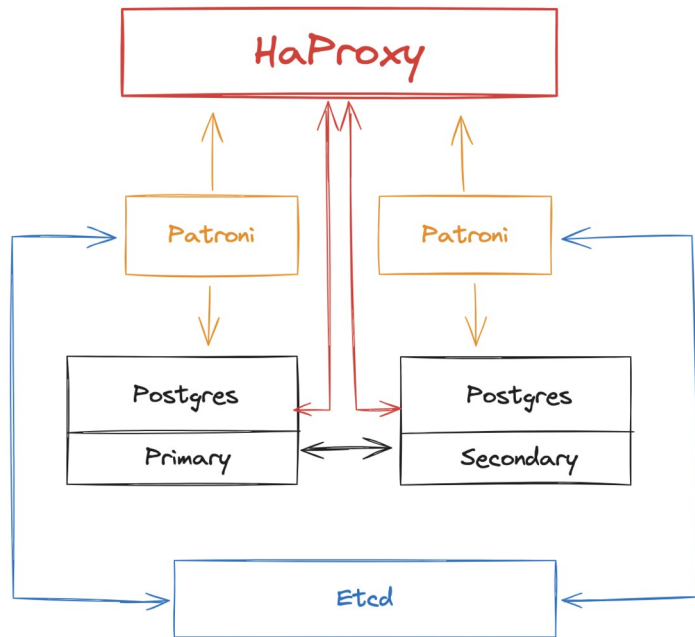
Tyk & Databases

- **Long-Term Data**
 - API definitions/Policies
 - Analytics/Usage Statistics
 - Dashboard User and Access Management
 - Audit Logs
- **Supported Databases**
 - PostgreSQL
 - MongoDB



PostgreSQL

- Combination of streaming replication, Patroni & Etcd for automated failover
- HAProxy for load balancing:
 - Continuous Synchronization by streaming write-ahead logs (WAL) from primary to standby nodes
 - HAProxy
 - Monitors health of servers via Patroni(Primary & Standby)
 - Distributes incoming database connections
 - Etcd acts as a distributed key-value store for configuration management
 - Patroni
 - Manages failover and PostgreSQL cluster management, monitors health of servers & checking status & connectivity
 - Orchestrates standby promotion if there are issues detected on primary server



MongoDB

MongoDB uses Replica Sets for high availability:

- Requires a minimum of three separate hosts for high availability
- All members of Replica Set must be able to communicate with each other
- Election occurs if Primary becomes inaccessible
- Members can have different priorities, to determine the new Primary
- A majority of Replica Set members are required for an election

